CLASSIFIED DOCUMENTATION

CAS: A-4757/EL/2221

1. Locator: [redacted]

2. Successor. cer: [redacted]

3. Project: [redacted]

4. Objective: [redacted]

5. [redacted]

6. [redacted]

7. [redacted]

8. [redacted]

9. [redacted]

10. [redacted]

11. [redacted]

Classified by: [redacted]

BUILDING TELENORS PRIVATE CLOUD

[redacted]



Aksel Skaar Leirvaag



Morten Neteland







CONFIDENTIAL

Project Nova

CLASSIFIED [redacted]

FAGBEVISNING.



TOP SECRET

DOC. BEH: [redacted]

ACCESS: CLASS B/B: [redacted]

SUBJECT: Telenor's Private Cloud / Project Nova

CLASSIFIED: [redacted]

CLASSIFICATION: TOP SECRET [redacted]

CLEARANCE: LEVEL [redacted]

LOCATION: [redacted]

DATE: [redacted]

DATUM: [redacted]

ORIGIN: [redacted]

STATUS: [redacted]

STATUS: [redacted]

STIFE: [redacted]



SURVEILLANCE PROFILE:

Strategic: Telenor's private cloud initiative to to create a secure and centralized infrastructure for critical [redacted] infrastructure for [redacted] operations, [redacted] operations [redacted] Operational security [redacted] Target: Aksel Skaar Leirvaag and Morten Neteland [redacted] operations of [redacted]

Threat: [redacted]

(Partial service records, medical history, and profile enclosed).

CLASSIFIED [redacted]

About Us



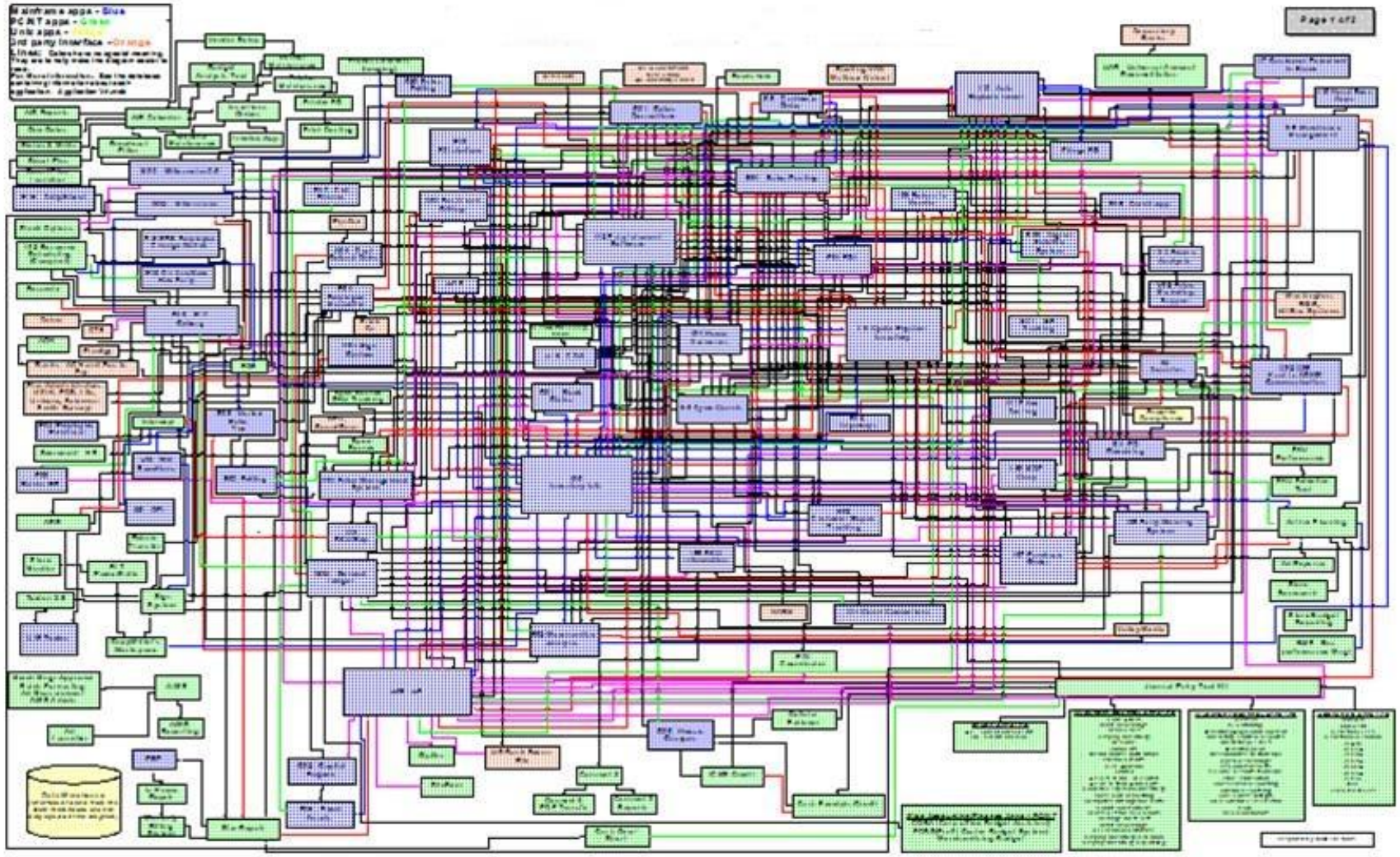
Morten Neteland
Platform Engineer
Telenor since 2021



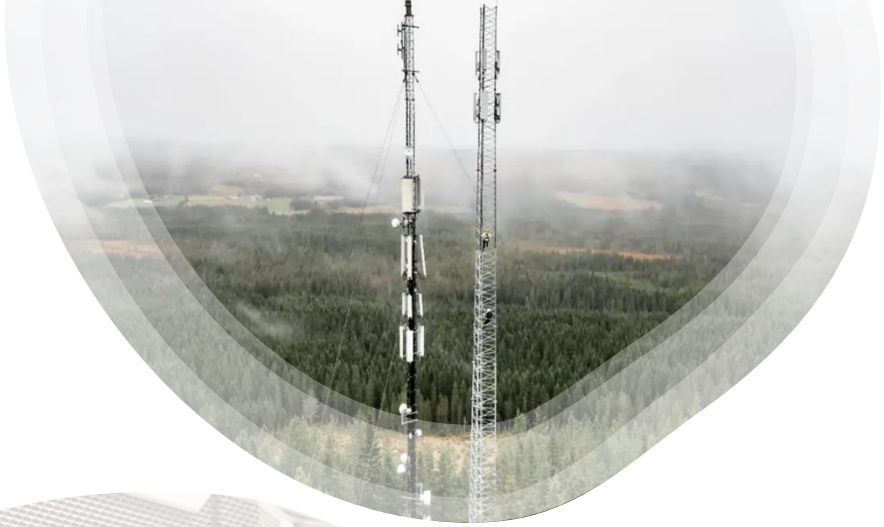
Aksel Skaar Leirvaag
Platform Engineer
Telenor since 2022

Sensitivity: Open

Today’s platform landscape



Sensitivity: Open





Why not public cloud?

- Platform and infrastructure for regulated workloads such as Telco services
- Hosting fundamental national functions, sometimes even supplying core networking to the hyperscalers.

Our background / toolbox

- Public Cloud, Azure and GCP, cloud babies
- Mainly focused on Kubernetes
- Built GitOps Kubernetes operators for self service



Sensitivity: Open



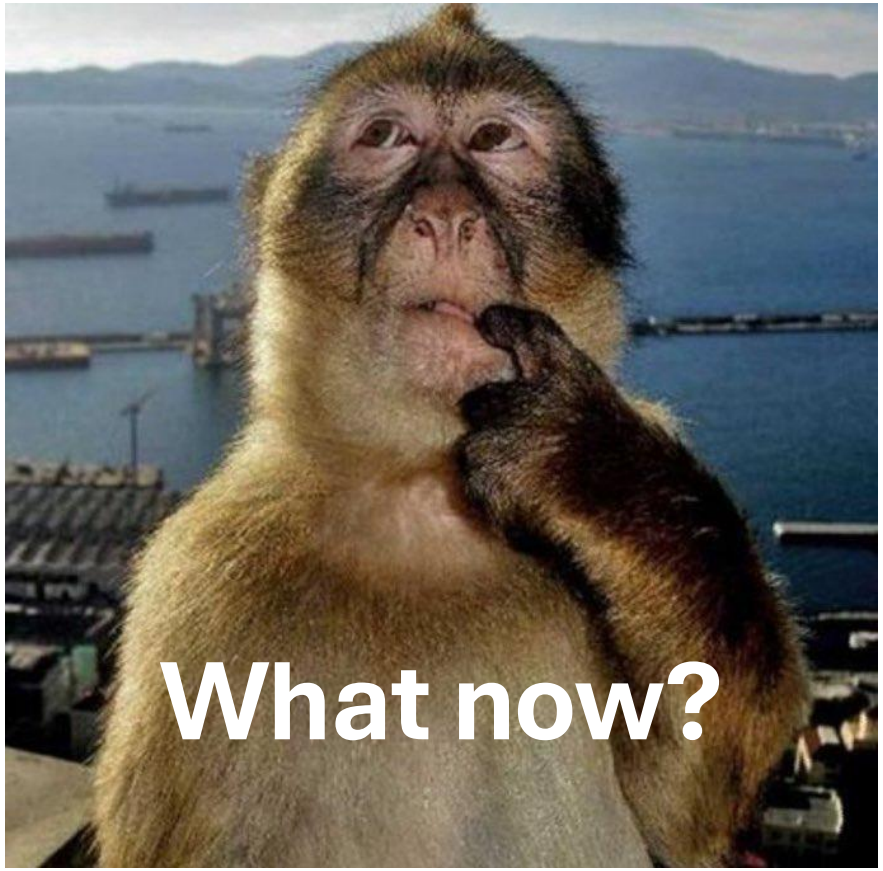
"Yes sir!"





"Can you deploy nova?"

Sensitivity: Open



Sensitivity: Open

Principles

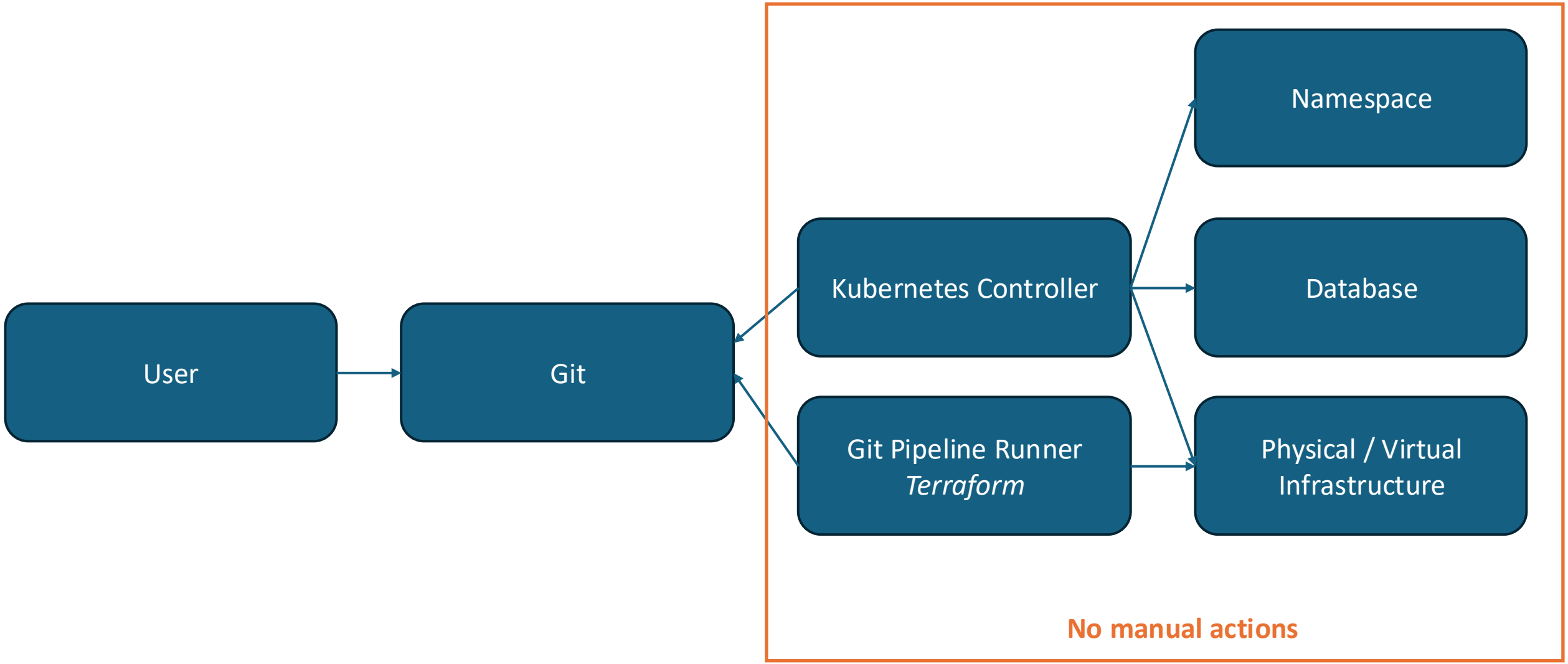
1. **Cloud Native**
2. Open Standards & CNCF Alignment
3. Vendor Neutral
4. Zero Trust, Secure by Default
5. Sustainable & Cost-Conscious
6. Kubernetes Resource Model (KRM)
7. Modular & Composable
8. GitOps & Declarative Everything
9. Observable by Design
10. Sovereign by Design

telenor



GitOps and Zero-Touch Provisioning

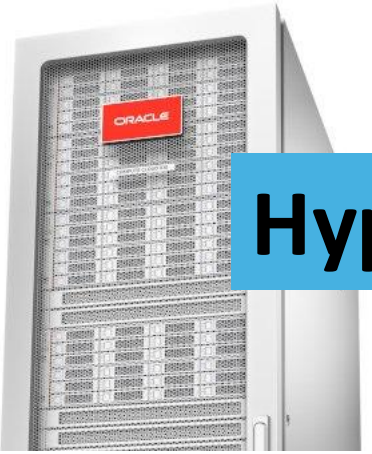
Our declarative vision



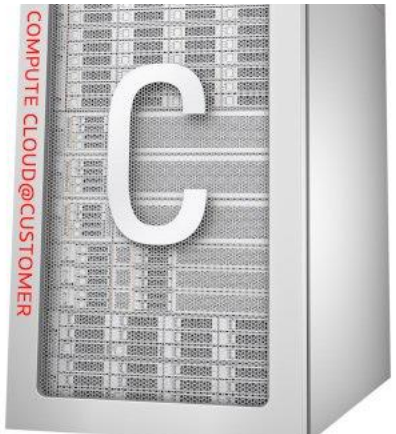
Sensitivity: Open

Hyperconverged Infrastructure

Azure Local



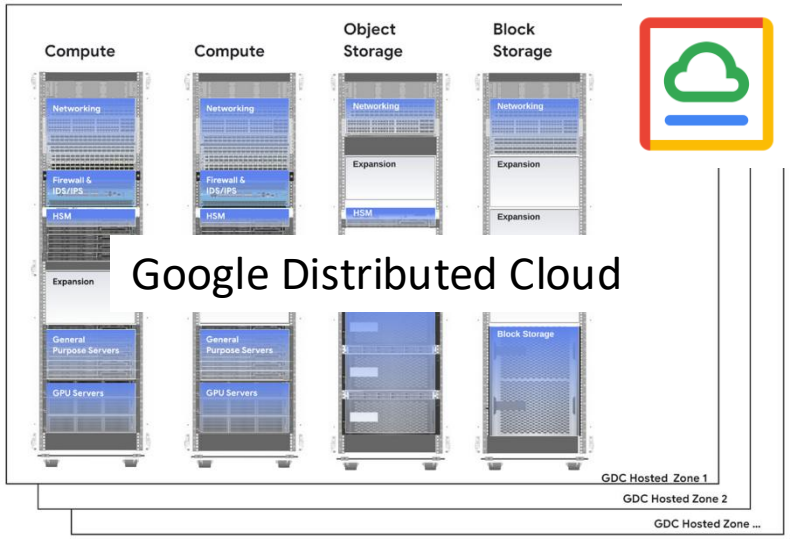
Oracle Cloud@Customer

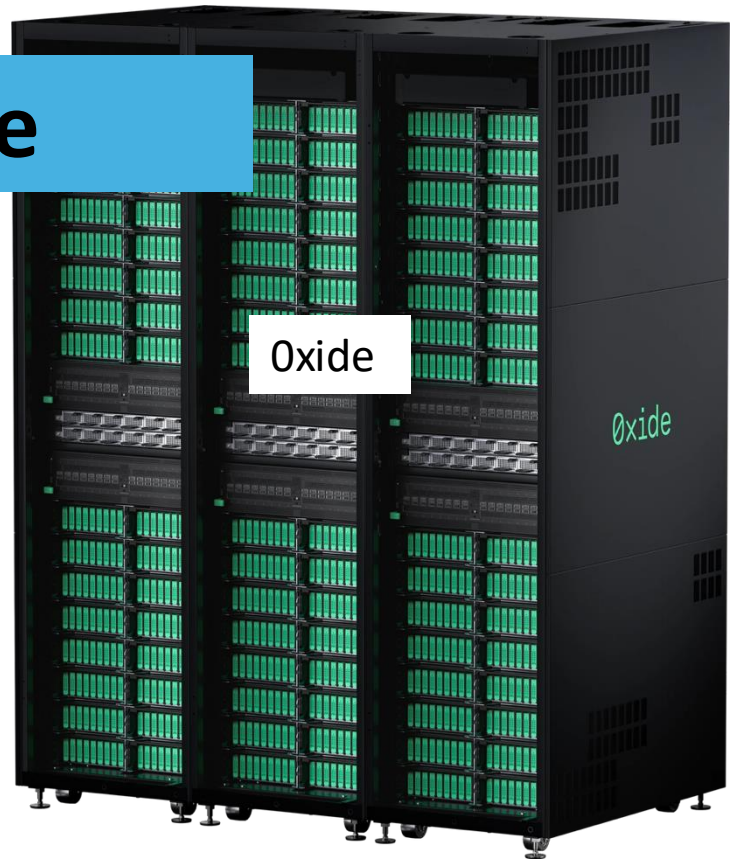




AWS Outposts







Sensitivity: Open

We got hardware!

What now?

Aksel and Morten



Compute

Black box, close to no insight



Network

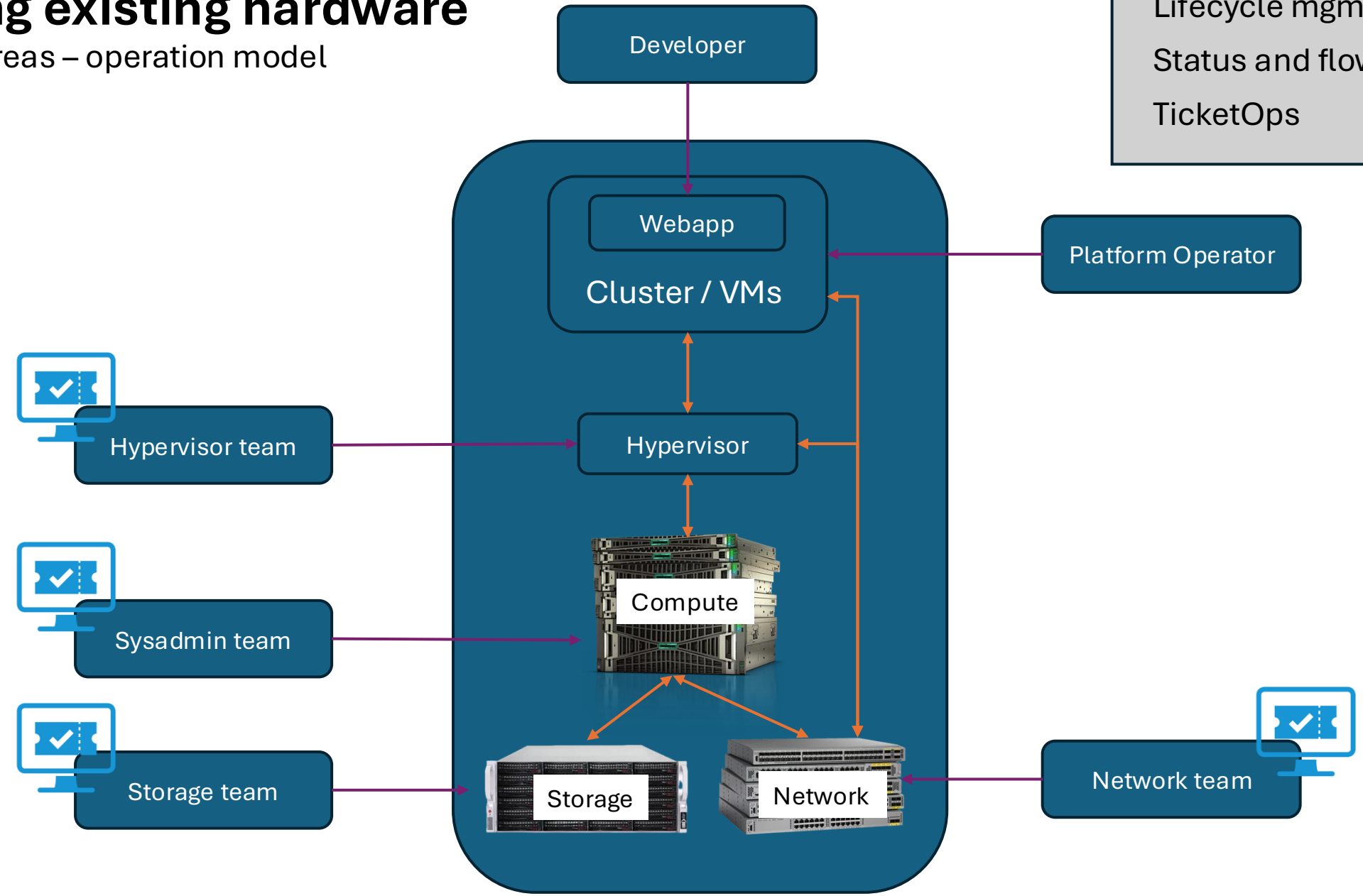


Storage

Sensitivity: Open

Utilizing existing hardware

Problem areas – operation model

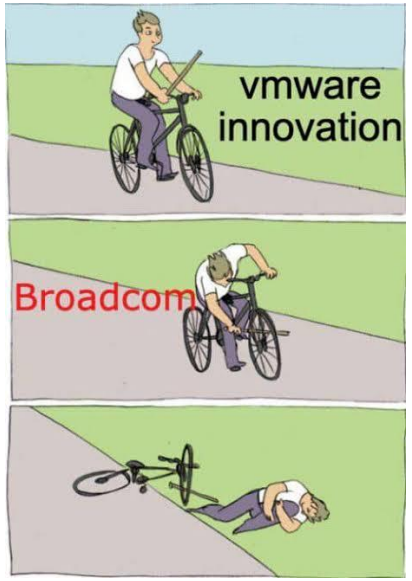


Sensitivity: Open

Utilizing existing hardware

Traditional hypervisors



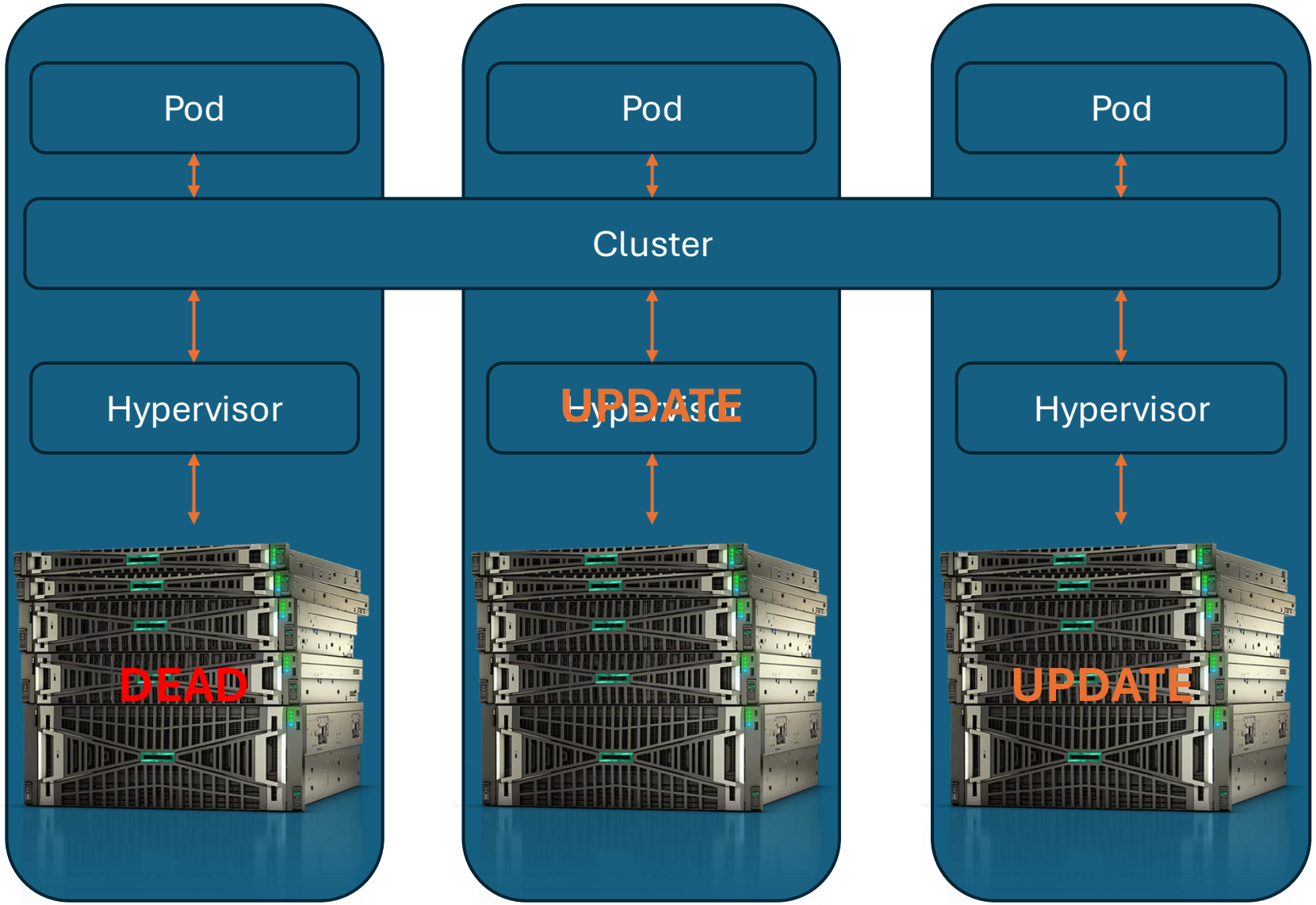


Sensitivity: Open

Utilizing existing hardware

Problem areas – aware scheduling

Propagation of status between compute layers

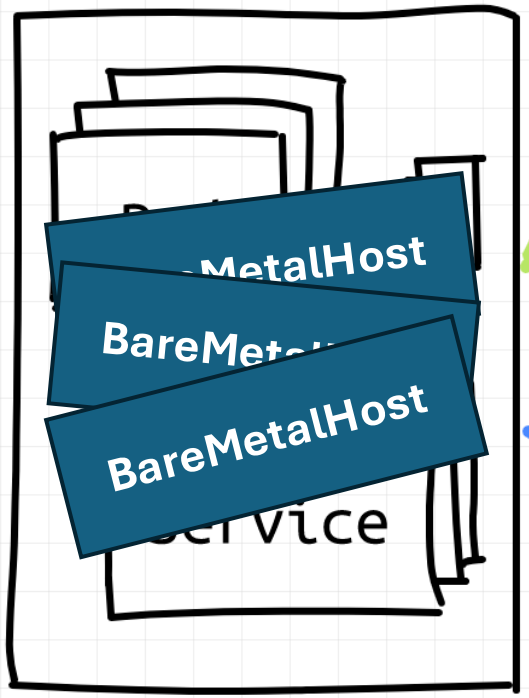


Sensitivity: Open

Kubernetes objects

Datacenter

~~System~~ Resources



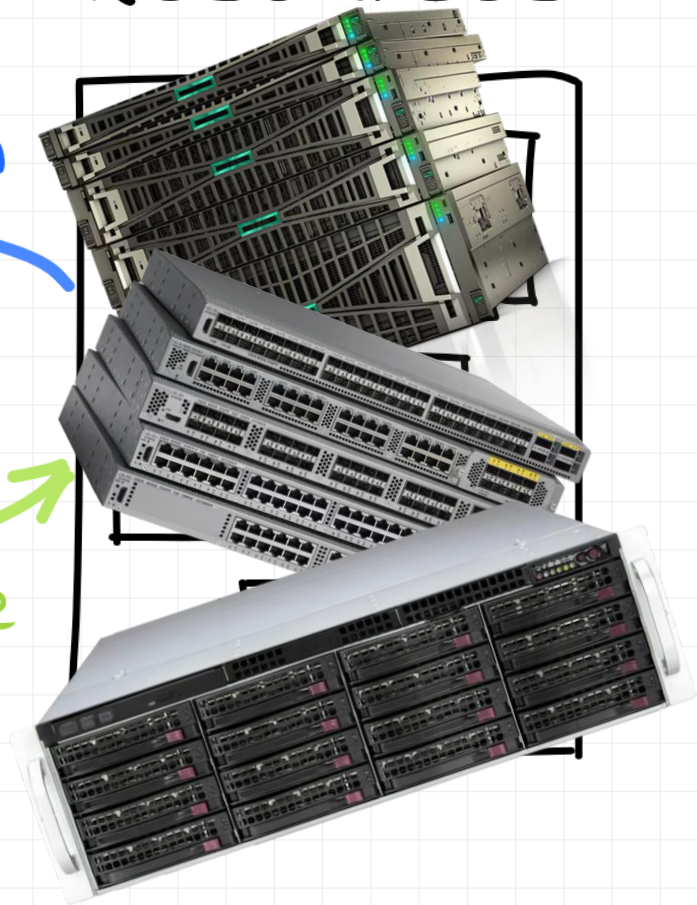
Update

watch

Controllers

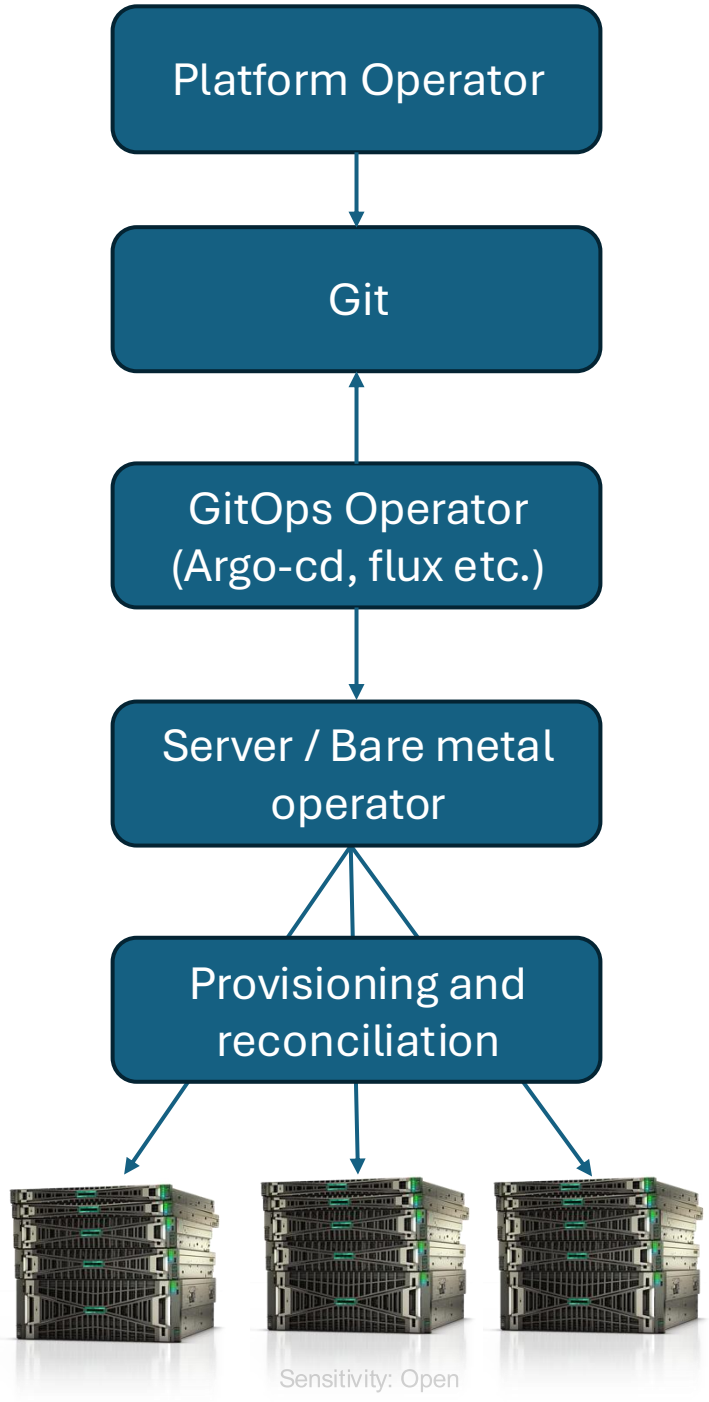
watch

Update



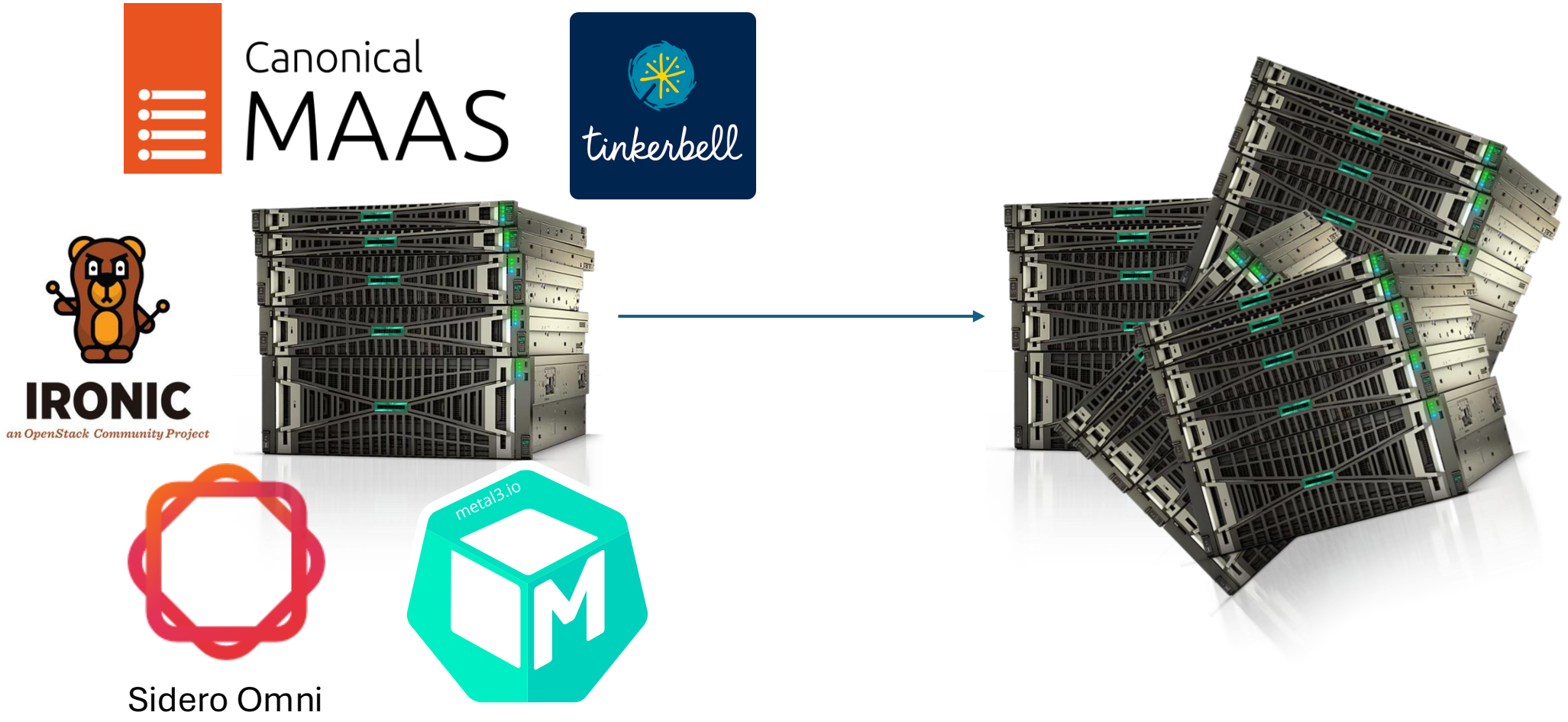
Sensitivity: Open

Desired state pattern



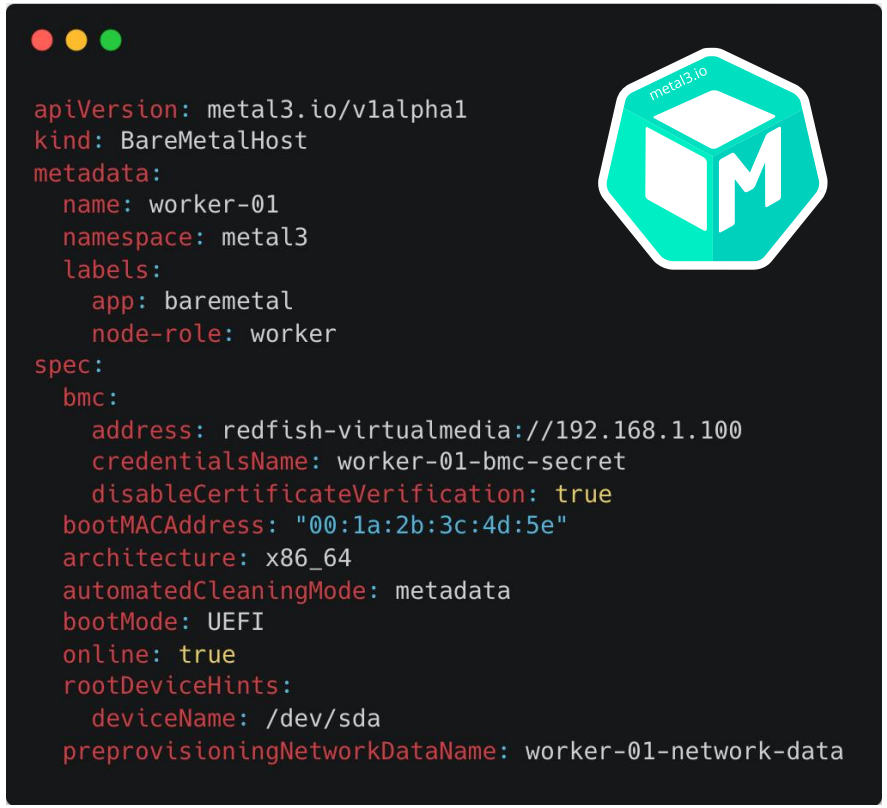
Sensitivity: Open

Bare metal provisioning

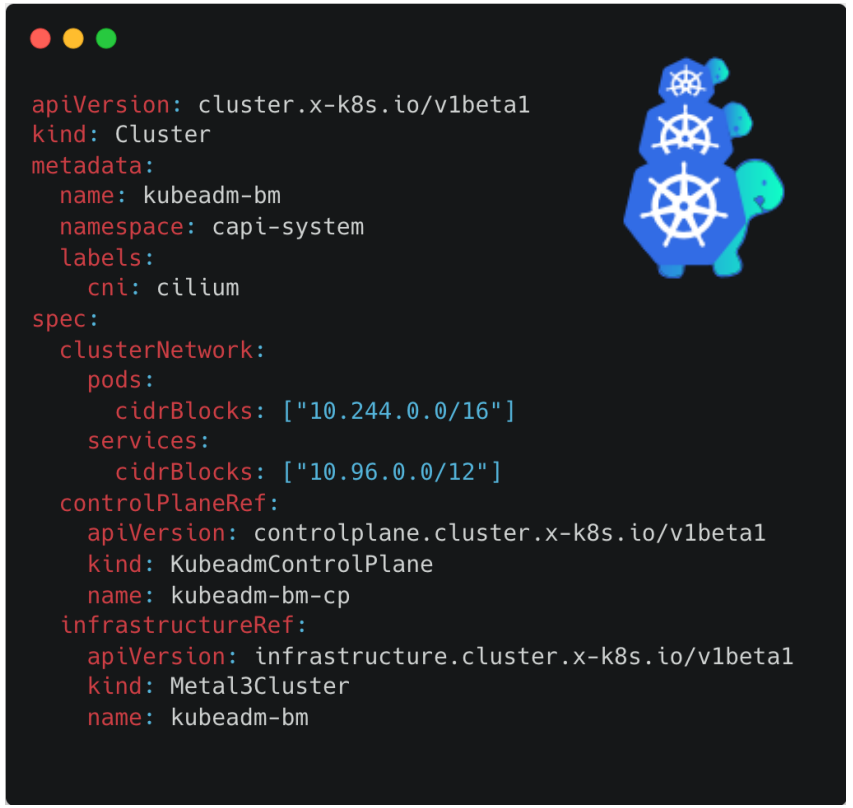


Sensitivity: Open

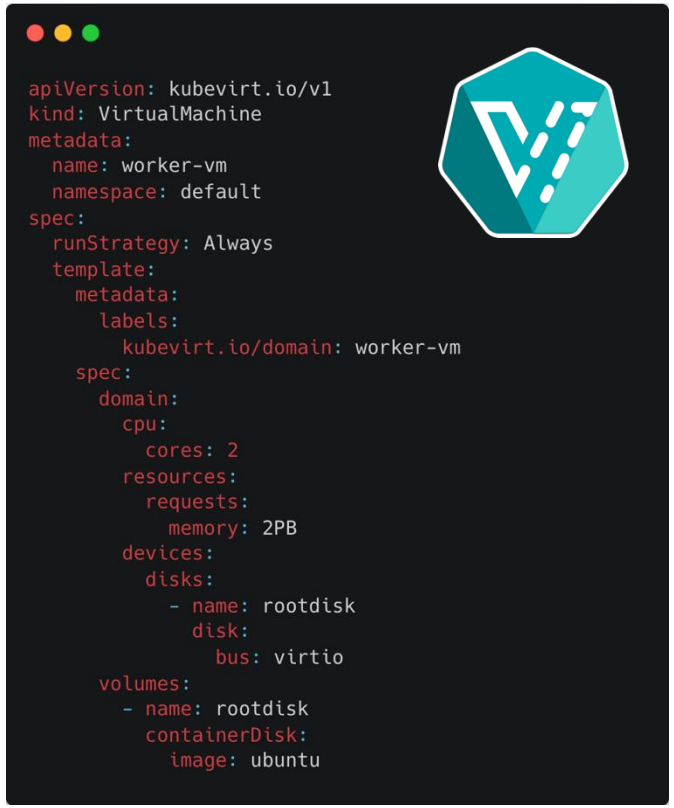
Our stack (mcapivirt)



Bare Metal provisioning



Cluster lifecycle management



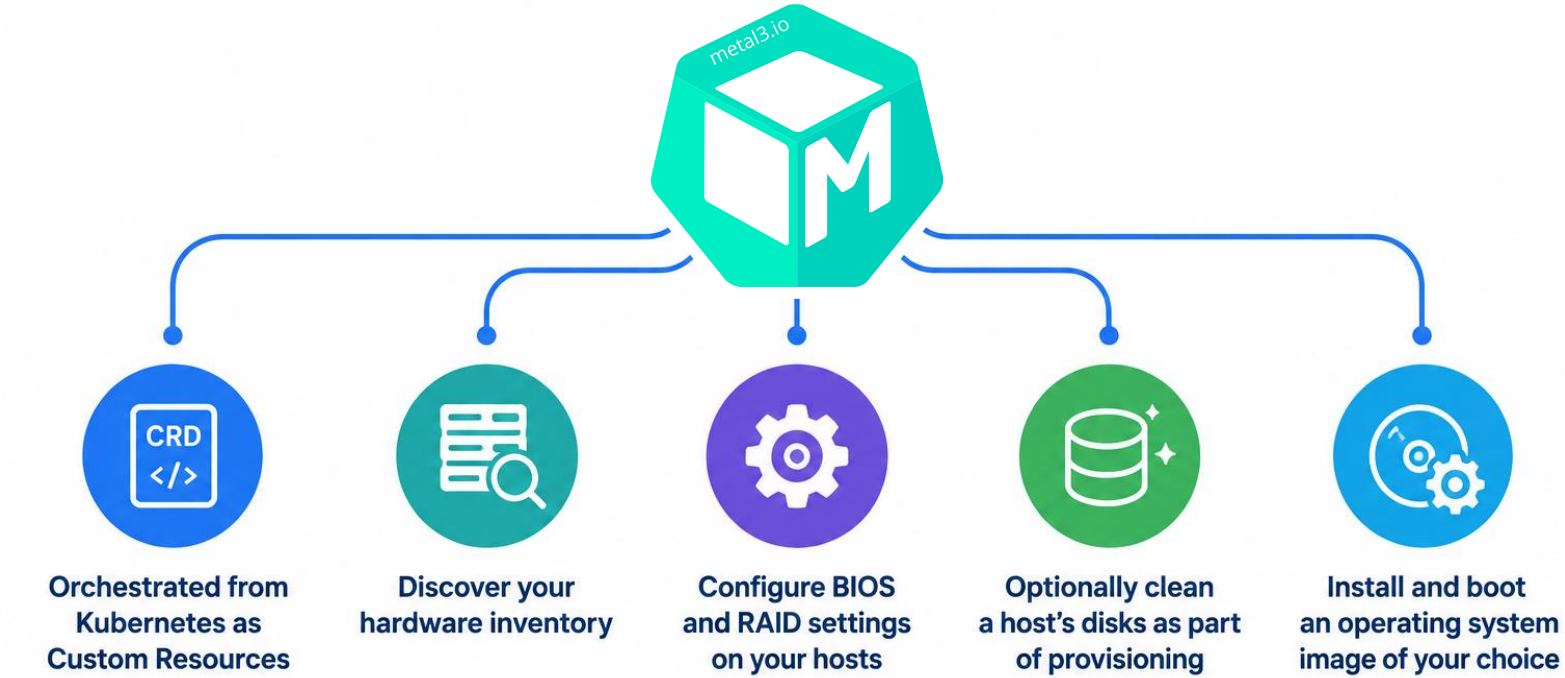
Kubernetes native VMs

Sensitivity: Open

Metal^3

Bare Metal provisioning

- CNCF Project
- Maintainers from RedHat and Ericsson
- Vendor neutral (HPE, Dell, Lenovo etc.)

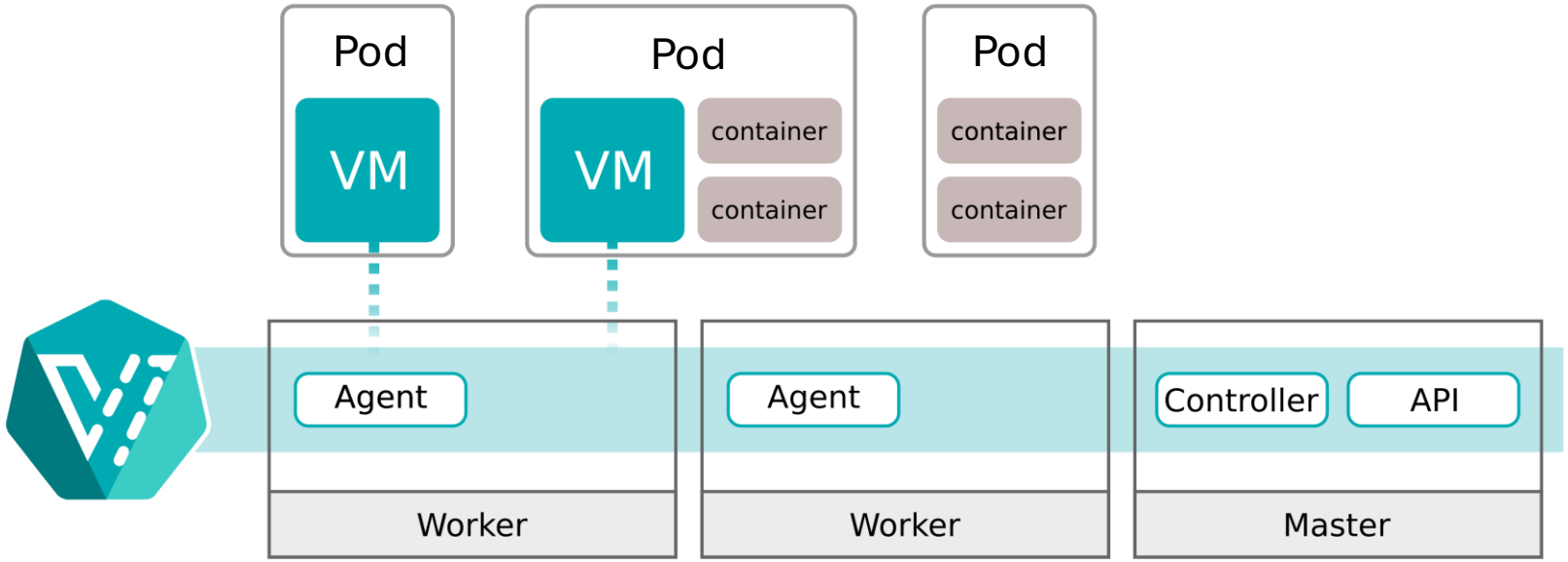


Sensitivity: Open

Kubevirt

Kubernetes native VMs

- CNCF Project
- Enabling pods to run KVM-based virtual machine inside
- Integrates nicely with cloud native container workloads



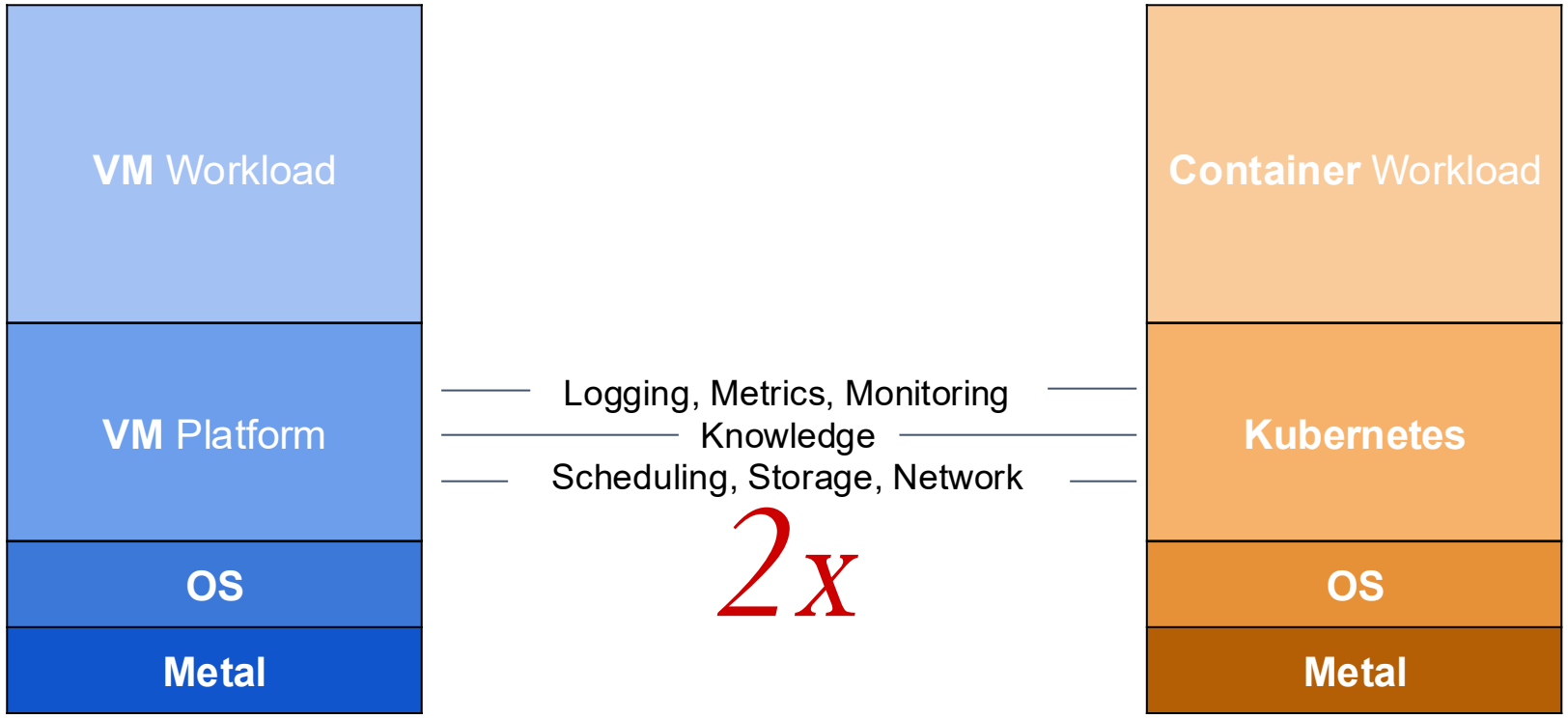

```yaml
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: worker-vm
  namespace: default
spec:
  runStrategy: Always
  template:
    metadata:
      labels:
        kubevirt.io/domain: worker-vm
    spec:
      domain:
        cpu:
          cores: 2
        resources:
          requests:
            memory: 2PB
        devices:
          disks:
            - name: rootdisk
              disk:
                bus: virtio
      volumes:
        - name: rootdisk
          containerDisk:
            image: ubuntu
```

Sensitivity: Open

Kubevirt

What does it solve?

Old Way … Multiple Workloads, Multiple Stacks

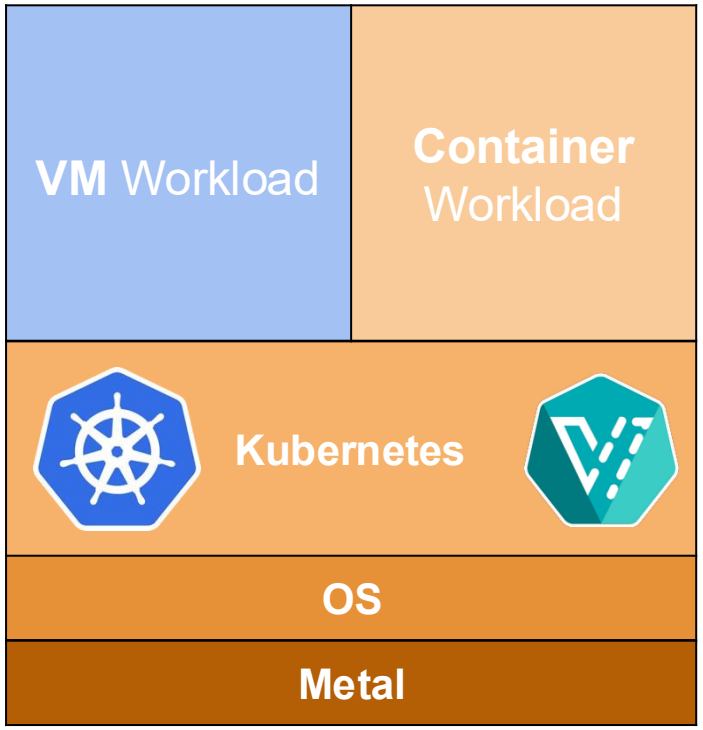


Sensitivity: Open

Kubevirt

What does it solve?

KubeVirt way… Multiple Workloads, One Stack



- Logging, Metrics, Monitoring
- Knowledge
- Scheduling, Storage, Network

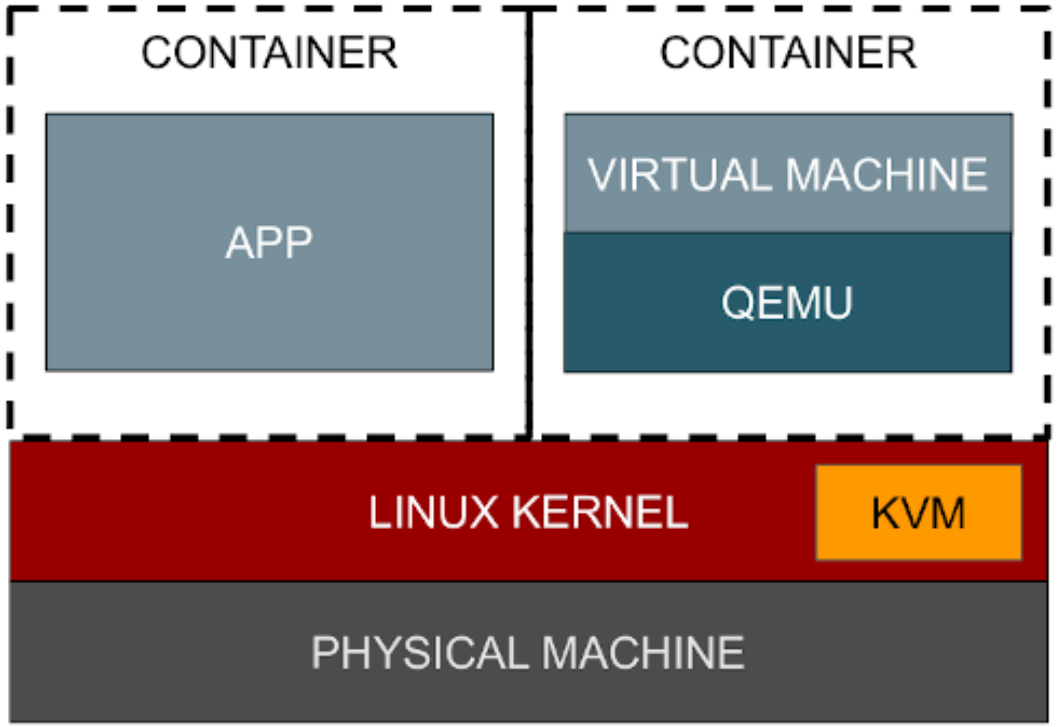
1 x

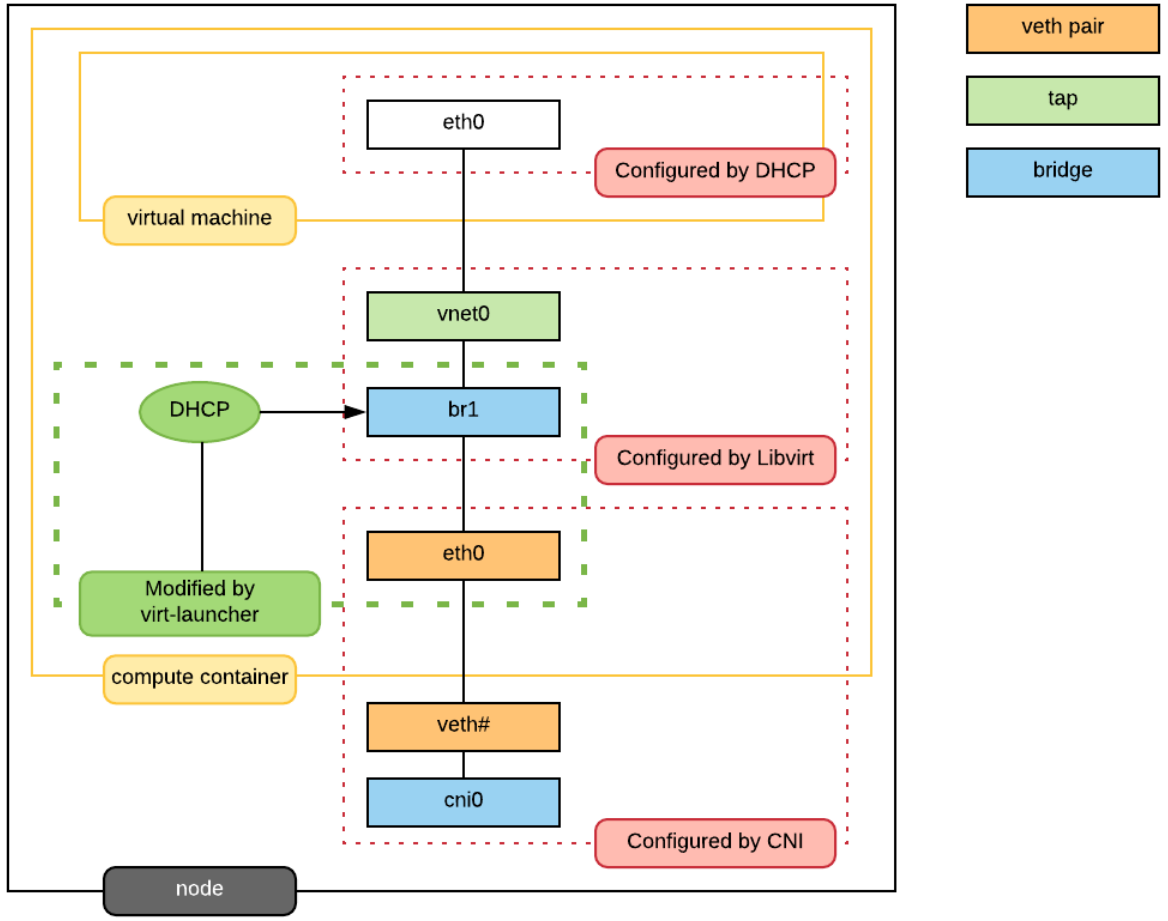
Sensitivity: Open

Kubevirt

Pod level

KubeVirt Razor (ideology): "If something is useful for Pods, we should not implement it only for VMs”

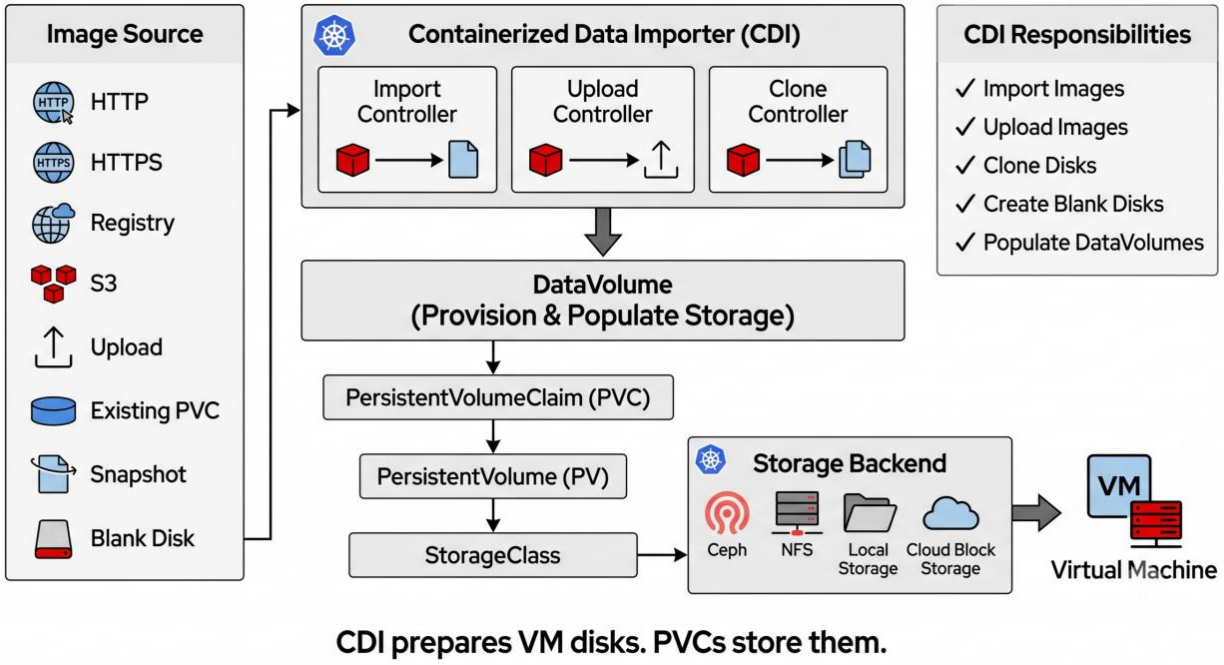




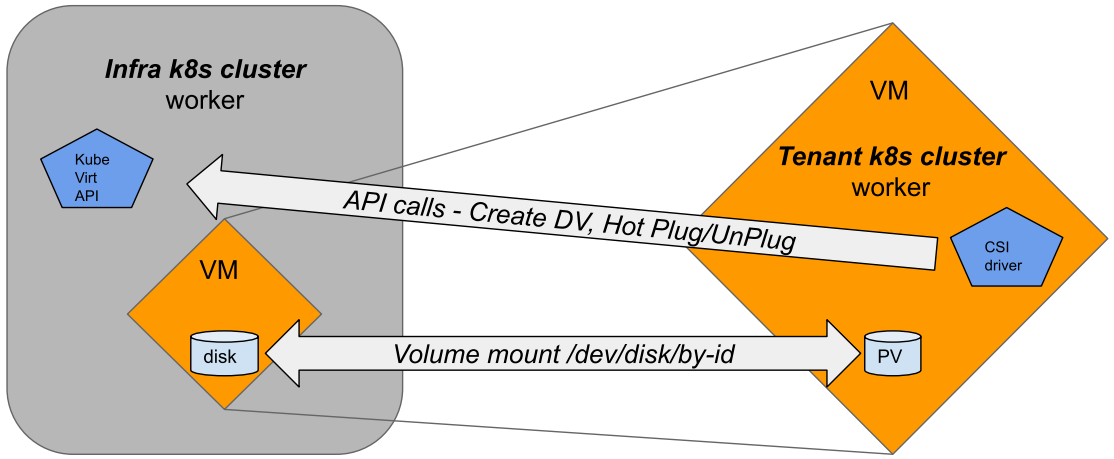
Sensitivity: Open

Kubevirt

Storage quirks



Using PVCs as Disks



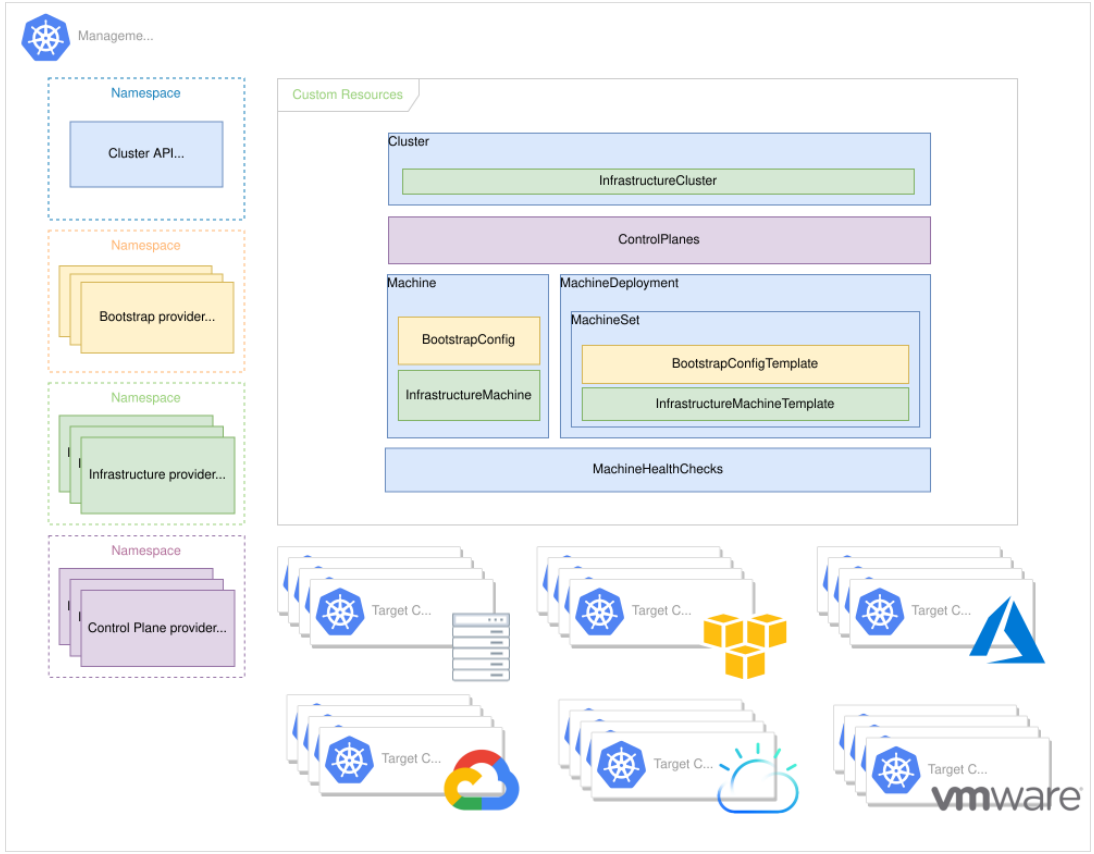
Supplying pods inside tenant cluster with infra cluster storage provider

Sensitivity: Open

Cluster API (CAPI)

Cluster lifecycle management

- Kubernetes subproject under sig-cluster-lifecycle
- Common implementation pattern across providers



Sensitivity: Open

Cluster API, simplified

Cluster

The cluster itself: endpoint, network, control plane

↓

MachineDeployment

Like a **Deployment**

↓

MachineSet

Like a **ReplicaSet**

↓

Machine

Like a **Pod**, but one node

→

Infrastructure

Which server: Metal3Machine

Bootstrap

How it joins: KubeadmConfig

Sensitivity: Open

CAPI lifecycle management



Cluster upgrades



Machine remediation



Firmware upgrades

Sensitivity: Open

Kubernetes node os image

OS image as a Containerfile

```
FROM quay.io/centos-bootc/centos-bootc:stream10

# Kubernetes and CRI-O minor versions move together
ARG K8S_VERSION=v1.34
ARG CRIO_VERSION=v1.34
#...
# Minimal node packages: runtime, kubelet, kubeadm and the tools kubelet expects
RUN dnf install -y --setopt=install_weak_deps=False \
      cri-o kubelet kubeadm cri-tools \
      iproute-tc iptables-nft conntrack-tools socat ethtool \
 && dnf clean all \
 && rm -rf /var/cache/dnf /var/log/dnf*

# Kernel modules and sysctls required for pod networking
RUN printf 'overlay\nbr_netfilter\n' > /usr/lib/modules-load.d/k8s.conf \
 && printf 'net.bridge.bridge-nf-call-iptables = 1\nnet.bridge.bridge-nf-call-
ip6tables = 1\nnet.ipv4.ip_forward = 1\n' \
      > /usr/lib/sysctl.d/99-k8s.conf

# Start the runtime and kubelet at boot; kubelet waits for kubeadm join/init to
write its config
RUN systemctl enable crio.service kubelet.service

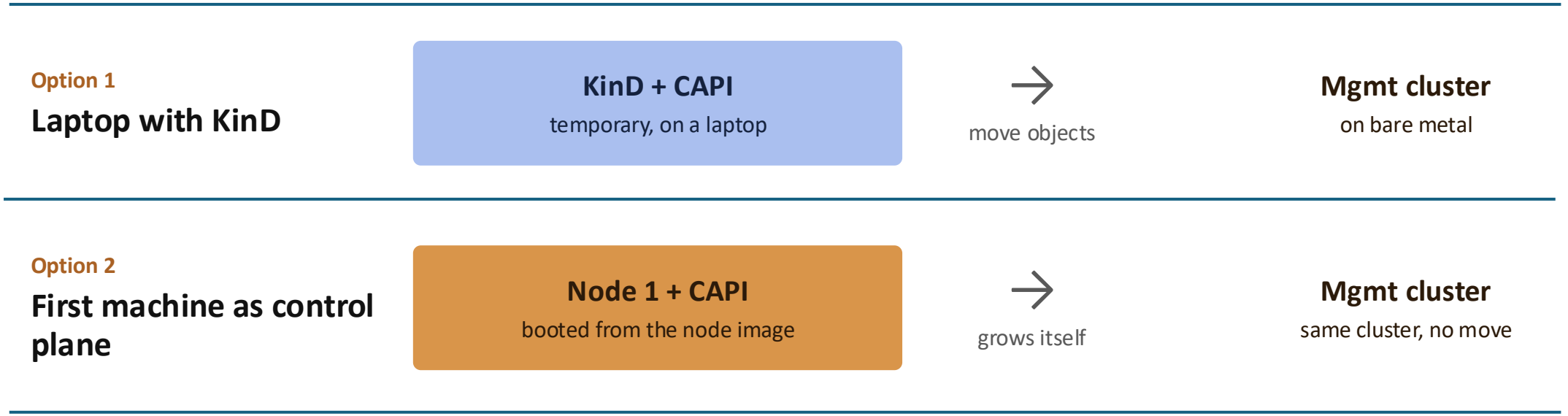
RUN bootc container lint
```

Sensitivity: Open

The bootstrap problem

Solving a chicken & egg problem

CAPI provisions clusters, but CAPI itself needs a cluster to run in...



Sensitivity: Open

The bootstrap ISO

One bootc image: the bootstrap host OS

Built online, boots offline

Kubernetes + first-boot tooling

kubeadm, cri-o and the bootstrap orchestrator

Container images

Every image the cluster will pull

Manifests and charts

Rendered from the repo

Node OS image + IPA ramdisk

Installed on every other host

Sensitivity: Open

From one ISO to a full cluster

01

Boot

Install the bootstrap ISO on one machine. A script sets up everything the initial cluster needs.

02

Seed



The CAPI stack comes up, and the machine becomes the first control-plane node.

03

Expand



ArgoCD applies the Cluster definition. CAPI provisions the remaining bare-metal machines.

04

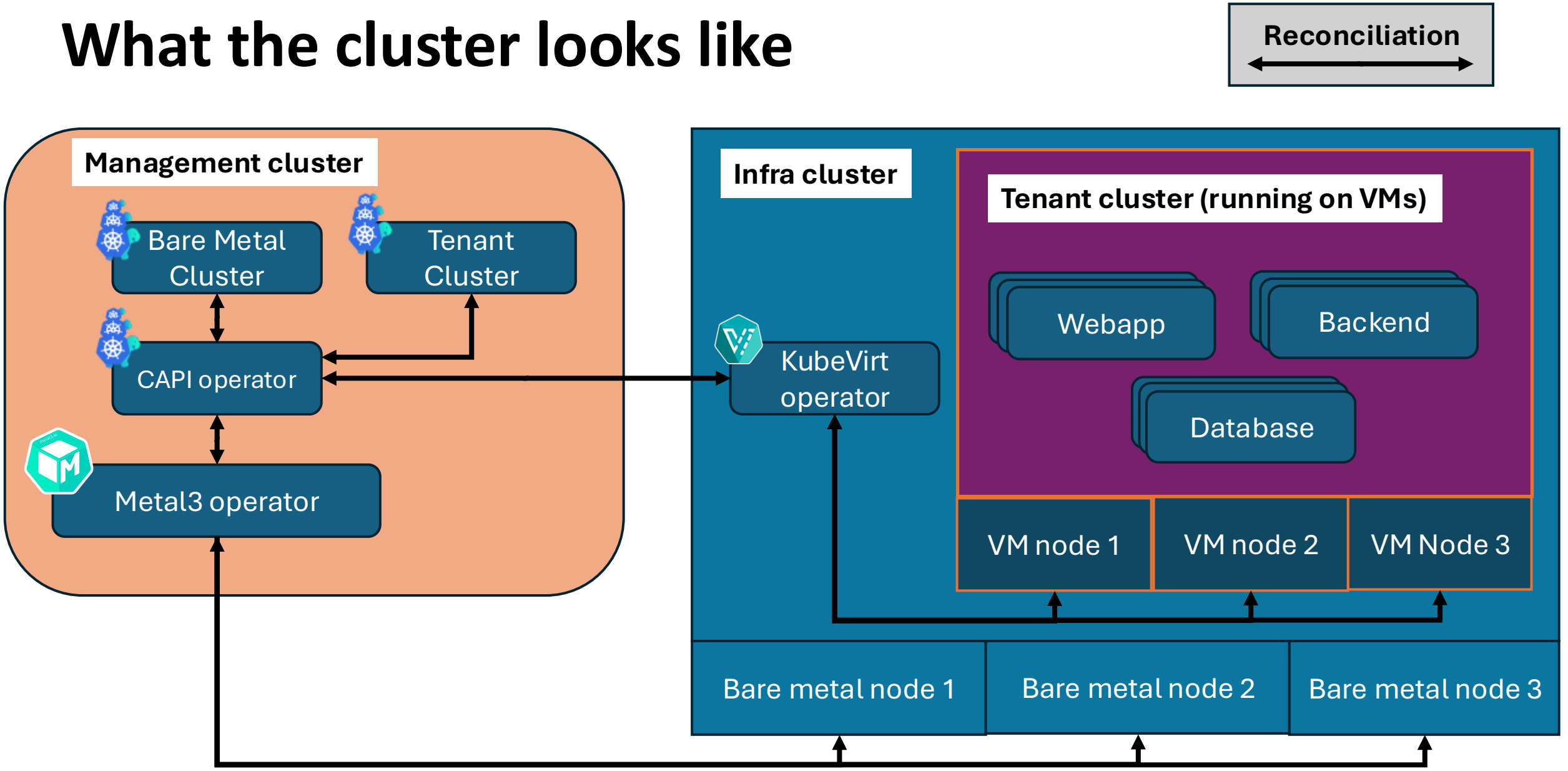
Adopt



The bootstrap machine is consumed as a normal CAPI machine. Nothing special is left behind.

Sensitivity: Open

What the cluster looks like



Sensitivity: Open

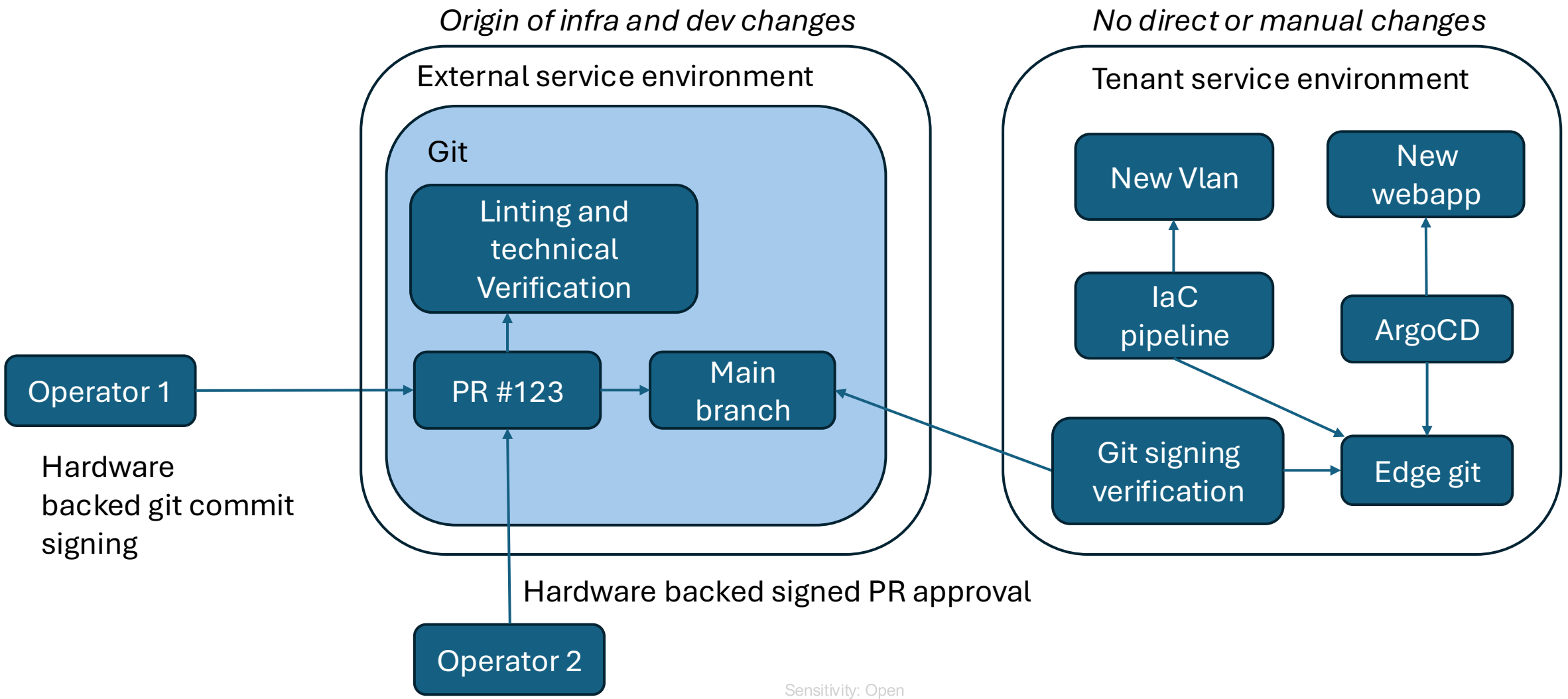
How we manage our platform



Sensitivity: Open

GitOps and Zero-Touch Provisioning

Signed artifact flow prototype



Sensitivity: Open

Authenticate with identity, not secrets

Humans
OIDC + YubiKey

Engineer runs kubectl → OIDC login via the IdP → YubiKey touch, phishing-resistant → **Token valid for minutes, audited**

Machines
Token exchange

Pod starts with a service account → Projected token, signed by cluster issuer → Exchanged against a trust binding → **Short-lived credential, audited**

Secure

Nothing to leak, share or rotate

Scalable

A new cluster adds one trusted issuer

Declarative

Revoking access = deleting a binding

Sensitivity: Open


Demo
Time!

Wrapping up

Achievements and uncertain terrain

- Holistic lifecycle management of bare metal to tenant cluster running on VMs
  - Aware of underlying infrastructure, all platform layers remediates on updates or errors.
- Sovereign software stack, vendor neutral and hardware independent. Enables us to deploy our entire stack at edge.
- Zero touch provisioning, from infrastructure to critical workloads
- Working on safely enabling multi tenant environments for our developers

Sensitivity: Open

Questions?



Morten Neteland
Platform Engineer



Aksel Skaar Leirvaag
Platform Engineer

Sensitivity: Open